The Holographic Alaya-Field Framework

The formation of accessible world structure

Complete Collected Volume

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Introduction

What makes a physical relation into part of a world, rather than a description imposed on it? HAFF follows the formation of persistent distinctions, their readable relations, temporal order and geometric response. The question is generative: which structures are produced by interaction, and which still enter through a physical dictionary or an initial boundary?

A and B begin with process-selected blocks and coupling-dependent coherence-decay relations. E asks how those distinctions become actual records. F shows how a strengthened recording protocol can identify an unknown order of events, while ordinary copying can remain order-blind. C connects acquired history to intervention. D uses a common conserved source to link modular constraints and physical response; G distinguishes internal execution from complete explanation.

These routes have different scopes. The decay-rate relation is not automatically space; transition responses require a separately identified transfer law. Recovered event order is not a generated time parameter, and the conditional linearized gravitational law still needs its field and entropy–area inputs. Finite return supplies a genuine common realization, not a substitute for those formation questions. The postscript brings the results back to cosmic origin and closure through an explicit formation-and-return conjecture.

Reading map

HAFF-A Physical coarse-graining selects persistent distinctions, with a coupling-dependent coherence-decay relation. Transfer responses and a spatial dictionary require further inputs; finite geometry certificates test this generative route.

HAFF-B Interaction-selected blocks, protected content and finite lifetimes establish which relations survive. Stable sectors are not automatically tensor factors or spatial sites.

HAFF-C Acquired history changes the interventions available to a body. The relation between functional agency and experience remains a separate explanatory question.

HAFF-D One conserved physical source enters modular and geometric response constraints. Conditional field-law synthesis, finite physical return and stronger thermal conjectures have distinct assumptions.

HAFF-E Actual instruments connect permissions to stable and public records. Public agreement requires more than redundancy to certify correctness.

HAFF-F Endpoint records identify an unknown exchange-event order under specified gates. Ordinary copying can fail this test; acquisition, renewal and time parametrization remain distinct.

HAFF-G Internal programs alter permissions and sources. Protocol-relative indistinguishability constrains complete explanation without creating a universal final boundary.

HAFF-PS Cosmic origin and closure return as a formation-and-renewal conjecture, alongside conditional finite-body participation bounds.